

Severity →										
Occurrence ↓	1.0	2.0	3.0	4.0	5.0	6.0	7.0	8.0	9.0	10.0
1.0	0.01	0.08	0.27	0.64	1.25	2.16	3.43	5.12	7.29	10.00
2.0	0.02	0.16	0.54	1.28	2.50	4.32	6.86	10.24	14.58	20.00
3.0	0.03	0.24	0.81	1.92	3.75	6.48	10.29	15.36	21.87	30.00
4.0	0.04	0.32	1.08	2.56	5.00	8.64	13.72	20.48	29.16	40.00
5.0	0.05	0.40	1.35	3.20	6.25	10.80	17.15	25.60	36.45	50.00
6.0	0.06	0.48	1.62	3.84	7.50	12.96	20.58	30.72	43.74	60.00
7.0	0.07	0.56	1.89	4.48	8.75	15.12	24.01	35.84	51.03	70.00
8.0	0.08	0.64	2.16	5.12	10.00	17.28	27.44	40.96	58.32	80.00
9.0	0.09	0.72	2.43	5.76	11.25	19.44	30.87	46.08	65.61	90.00
10.0	0.10	0.80	2.70	6.40	12.50	21.60	34.30	51.20	72.90	100.00
Detection →	1.00	2.0	3.0	4.0	5.0	6.0	7.0	8.0	9.0	10.0

$$Formula: RPN = [(Severity^2) \times Occurrence \times Detection] / 100$$

Risk Priority Number (RPN) Scale & Interpretation		
Low Risk	0.00 – 1.24	Tolerable with monitoring
Moderate Risk	1.25 – 5.49	Action required in timely manner
High Risk	5.50 – 10.00	Immediate action required
Severity	1 = Negligible (No Injury, No Damage) 2 = Very Minor (First Aid Only) 3 = Minor (Medical, No Lost Time) 4 = Moderate (Recordable Injury, Restricted Duty) 5 = Significant (Lost-Time Injury) 6 = Serious (Permanent Minor Impairment) 7 = Major (Permanent Disability) 8 = Critical (Life-Threatening Injury) 9 = Catastrophic (Fatality Possible) 10 = Extreme (Certain Fatality)	
Occurrence	1 = Rare (Random to Never) 2 = Very Uncommon (About once per Year) 3 = Uncommon (Few times per Year) 4 = Infrequent (Several times per Year) 5 = Occasional (About once per Month) 6 = Recurring (Every few Weeks) 7 = Frequent (About once per Week) 8 = Very Frequent (About once a Day) 9 = Routine (few times a day) 10 = Constant (Multiple times per Shift)	
Detection	1 = Obvious (noticeable even if distracted) 2 = Obviously Visible (Almost always noticed) 3 = Clearly Visible (Usually noticed) 4 = Visible (Often noticed) 5 = Slightly unnoticeable (overlooked) 6 = Unnoticeable (Missed if unfocused or unfamiliar) 7 = Difficult to see (Rarely noticed) 8 = Semi-hidden (partially covered or hidden) 9 = Hidden (Mostly covered or hiding) 10 = Near Invisibly (Completely covered or darkness)	

Post-Control Risk Review:

Controls Added:

Control Types: Elimination / Substitution / Engineering / Administrative / PPE

New Severity Rating (S):

New Occurrence Rating (O):

New Detection Rating (D):

New FMEA Total:

Action Status: Acceptable / Monitor / Improve Further / Unacceptable

Severity-Weighted FMEA Model (Murphy, 2025)

Developed for Mining Risk Prioritization

